

DWM : UT2 QB

- Q.1] Define Cluster Analysis. Give the application of cluster analysis.
- Q.2] Differentiate between Classification and Clustering algorithms.
- Q.3] Explain k-Means Algorithm in detail with a suitable example.
- Q.4] Describe in detail Market Basket Analysis.
- Q.5] Differentiate between Apriori Algorithm and FP-Growth Algorithm. Chp5
- Q.6] What is Web mining? Explain techniques of web Mining. Chp6
- Q.7] Explain Apriori Algorithm with flowchart. Chp5
- Q.8] Explain the FP-Growth Algorithm with its advantages and disadvantages. Chp5
- Q.9] Compare and contrast data mining and web mining. Chp6.
- Q.10] What is Web structure mining? Chp6
- Q.11] Using k-means Clustering cluster the foll. data into two clusters and show each step.

(2, 4, 10, 12, 3, 20, 30, 11, 25)

- Q.12] Demonstrate Multidimensional and Multilevel Association Rule Mining with a suitable example.

- Q.13] Given the following data, apply the apriori algo. Given support threshold = 50%, Confidence = 60%.

Transaction	List of Items	Chp5
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T ₁	11, 12, 13	
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T ₂	12, 13, 14	
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T ₃	14, 15	
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T ₄	11, 12, 14	
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T ₅	11, 12, 13, 15	
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T ₆	11, 12, 13, 14	
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Ex. 4.4.1 : Using k-means clustering, cluster the following data into two clusters and show each step. Q B 2

{2, 4, 10, 12, 3, 20, 30, 11, 25}

✓ **Soln. :**

- **Step 1 : Randomly partition the given data set into two clusters and find the cluster mean.**

$$k_1 = \{2, 3\} \rightarrow m_1 = 2.5$$

$$k_2 = \{4, 10, 12, 20, 30, 11, 25\} \rightarrow m_2 = 16$$

- **Step 2 : Reassign the data points to each cluster and find again the cluster mean.**

$$k_1 = \{2, 3, 4\} \rightarrow m_1 = 3$$

$$k_2 = \{10, 12, 20, 30, 11, 25\} \rightarrow m_2 = 18$$

- **Step 3 : Reassign and find cluster mean.**

$$k_1 = \{2, 3, 4, 10\} \rightarrow m_1 = 4.75$$

$$k_2 = \{12, 20, 30, 11, 25\} \rightarrow m_2 = 19.6$$

- **Step 4 : Reassign and find cluster mean.**

$$k_1 = \{2, 3, 4, 10, 11, 12\} \rightarrow m_1 = 7$$

$$k_2 = \{20, 30, 25\} \rightarrow m_2 = 25$$

- **Step 5 : Reassign and find cluster mean.**

$$k_1 = \{2, 3, 4, 10, 11, 12\} \rightarrow m_1 = 7$$

$$k_2 = \{20, 30, 25\} \rightarrow m_2 = 25$$

- **Step 6 : Stop. The clusters in step 4 and 5 are same.**

Final answer: $k_1 = \{2, 3, 4, 10, 11, 12\}$ and $k_2 = \{20, 30, 25\}$

4) Describe in detail Market Basket Analysis.

Ans:- Market Basket Analysis (MBA) is a data mining technique used to find relationships or patterns between items that are frequently bought together by customers.

1) It helps businesses understand which products are often purchased together, so they can make better marketing, pricing and placement decisions.

2) It is also known as Association Rule Learning and it is commonly used in supermarkets, online shopping and retail stores.

3) The main goal of Market Basket Analysis is to study customers' buying habits and find connections between different products.

For example: If many customers who buy bread also buy butter, the store can place them close together or offer a combo discount.

Key Term in Market Basket Analysis:

- 1) Itemset :- A collection of one or more items.
Example : {Milk, Bread}, {Soap, Shampoo}
- 2) Confidence : It shows how often items are bought together.
- 3) Support : It tells how often an item or itemset appears in all transactions.
- 4) Lift : It measures how strong the relationship is between items.

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Applications of MBA:

- 1) Retail and supermarkets.
- 2) E-commerce websites.
- 3) Banking
- 4) Telecommunication
- 5) Healthcare

- The association rule has three measures that express the degree of confidence in the rule. They are Support, Confidence, and Lift.
- The **support** is the number of transactions that include items in the {A} and {B} parts of the rule as a percentage of the total number of transactions. It is a measure of how frequently the collection of items occur together as a percentage of all transactions. Also called the **occurrence frequency, frequency, support count, or count**.

Support ($A \rightarrow B$)

$$= \frac{\text{Number of transactions containing both A and B}}{\text{Total number of transactions}}$$

- The **confidence** of the rule is the ratio of the number of transactions that include all items in {B} as well as the number of transactions that include all items in {A} to the number of transactions that include all items in {A}.

Confidence ($A \rightarrow B$)

$$= \frac{\text{Number of transactions containing both A and B}}{\text{Number of transactions containing A}}$$

- The **lift or lift ratio** is the ratio of confidence to expected confidence. Expected confidence is the confidence divided by the frequency of B. The Lift tells us how much better a rule is at predicting the result than just assuming the result in the first place. Greater lift values indicate stronger associations.

Lift ($A \rightarrow B$)

$$= \frac{\frac{\text{Number of transactions containing both A and B}}{\text{Number of transactions containing A}}}{\frac{\text{Number of transactions containing B}}{\text{Total number of transactions}}}$$

- **Example :** Consider there are nine baskets containing varying combinations of milk, cheese, apples, and bananas.

Basket	Product 1	Product 2	Product 3
1	Milk	Cheese	
2	Milk	Apples	Cheese
3	Apples	Banana	
4	Milk	Cheese	
5	Apples	Banana	
6	Milk	Cheese	Banana
7	Milk	Cheese	
8	Cheese	Banana	
9	Cheese	Milk	

Web
Structure
Mining

Web
Content
Mining

Web
Usage
Mining

Fig: Web Mining Techniques

1) Web Content Mining:-

- It focuses on extracting useful information from the contents of web pages such as text, images, audio, video and metadata.
- It helps in identifying the relevant information from large web data.
- Example:- Extracting product details from e-commerce web sites.

2) Web Usage Mining:-

- It focuses on analyzing user behavior on websites using server logs, cookies and clickstream data.
- It helps in understanding how user navigate and interact with a website.
- Example:- Analyzing user click patterns to improve website design or recommend products.

3) Web Structure Mining:-

- Web structure mining is a type of web mining that focuses on analyzing the structure of hyperlinks between web pages.
- It helps to understand the relationship between

web pages and how they are connected.

- In simple words, It is about understanding the connections and relationships between different websites or pages, rather than the content of the page.
- Example:- Google's PageRank algorithm uses structure mining to rank web pages.

Techniques of web structure mining:-

- 1) Analyzing Hyperlinks.
- 2) Graph Theory methods
- 3) Social Network Analysis
- 4) PageRank Algorithm
- 5) Link Analysis
- 6) Community detection

Applications:-

- 1) Search Engine Optimization (SEO)
- 2) Website Navigation Improvement
- 3) Recommendation System
- 4) Fraud Detection

16/10/25 *Answers:—

Q. Differentiate between Apriori Algorithm and FP-Growth Algorithm

→

ParametersApriori Algo.FP-Growth Algo.

Storage structure.

Array based

Tree based

Search Type

Breadth first search

Depth first search

No. of database

multiple scans ($k+1$)

Only twice scans

Scans

for generating candidate sets.

for constructing frequent

Memory

Utilization

Large Memory Space

pattern tree.

(Candidate generation)

Less memory space

(No candidate gen.)

Run time

More time

Less time

Database

Sparse/dense datasets

Large & Medium datasets

Approach

Apriori utilizes a Level-wise approach, where it generates patterns containing 1 item, 2 items then 3 items & so on.

FP Growth utilizes a pattern-growth approach, it only considers patterns actually existing in the

Performance

Perform poorly on large datasets.

Works well on large datasets

Cost Computational

High

Low

Q. Explain Apriori Algorithm with eg. flowchart.

- Apriori Algorithm is a popular algorithm used for association rule mining in data mining.
- Apriori Algorithm was first algorithm that was proposed for frequent itemset mining.
- It was later in 1994 improved by Rakesh Agrawal and Ramkrishnan Srikant and came to be known as Apriori.
- Apriori algorithm is a sequence of steps to be followed to find the most frequent itemset in the given database.
- This algorithm uses two steps "join" & "prune" to reduce the search space.

i) Join step:

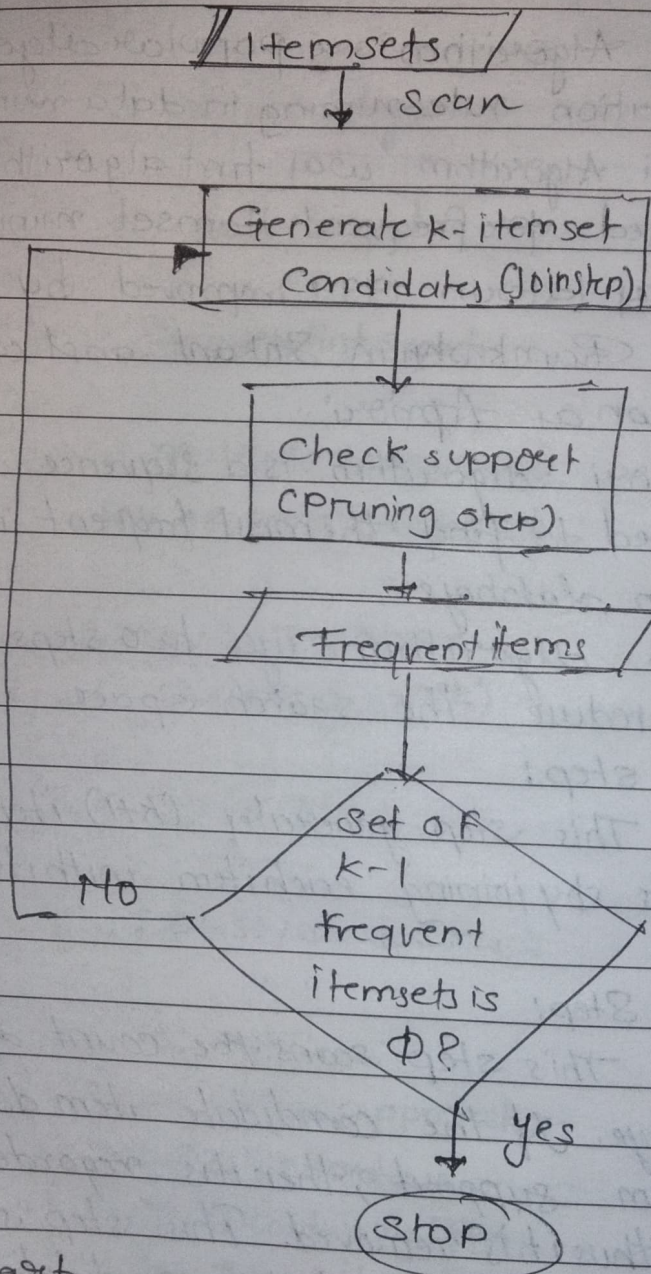
This step generates $(k+1)$ itemsets from k itemsets by joining each item with itself.

ii) Prune Step:

This step scans the count of each item in the database. If the candidate item does not meet minimum support, then it is regarded as infrequent and thus it is removed. This step is performed to reduce the size of the candidate itemsets.

- This data mining technique follows the join and the prune steps iteratively until the most frequent itemset is achieved.

Itemsets →



1. Start
2. Scan the datasets to generate Candidate (itemsets).
3. Generate k-itemset candidates (Joinstep):
The algorithm generates Candidate itemsets of size k by joining frequent itemsets of size k-1

4. Check Support (prune step):

The algorithm checks the support of these Candidate itemsets and prunes those that don't meet the minimum support threshold.

5. Frequent items: The remaining itemsets after pruning are considered frequent items.

6. Check if set of $k-1$ frequent itemsets is $\emptyset$?

• If yes, the process stops.

• If no, the algorithm continues to generate the next set of Candidate itemset.

Q7] Given the foll. data, apply the apriori algorithm.
Given support threshold = 50%, confidence = 60%.

Transaction	List of Items
T_1	I_1, I_2, I_3
T_2	I_2, I_3, I_4
T_3	I_4, I_5
T_4	I_1, I_2, I_4
T_5	I_1, I_2, I_3, I_5
T_6	I_1, I_2, I_3, I_4

→ Support threshold = 50%.

$$\begin{aligned}\therefore \text{min-sup} &= 0.5 \times \text{no. of transaction} \\ &= 0.5 \times 6 \\ &= 3\end{aligned}$$

Thus, min-sup = 3

Step 1: Count of each Itemset (C1) by scanning the database.

Itemset	Count
{I1}	4
{I2}	5
{I3}	4
{I4}	4
{I5}	2 < 3

Step 2: Prune Step (L1): C1 shows that I5 itemset does not meet min-sup³, thus it is deleted only.

Itemset	Count
{I1}	4
{I2}	5
{I3}	4
{I4}	4

Step 3: Join Step: Form C2 from L1 and L1.

Itemset	Count
{I1, I2}	4
{I1, I3}	3
{I1, I4}	2 < 3
{I2, I3}	4
{I2, I4}	3
{I3, I4}	2 < 3

Step 4: Prune Step (L2): C2 shows that itemset {I2, I4} & {I3, I4} does not meet min-sup, thus it is discarded.

Itemset	Count
{I1, I2}	4
{I1, I3}	3
{I2, I3}	4
{I2, I4}	3

Step 5: Join Step: Form C3 from L2 using L2 and L2.

Itemset	Count
{I1, I2, I3}	3
{I1, I2, I4}	2 < 3
{I1, I3, I4}	1 < 3
{I2, I3, I4}	2 < 3

Step 6: Prune Step (L3): C3 shows that itemsets {I1, I2, I4} & {I1, I3, I4} & {I2, I3, I4} does not meet min-sup, thus it is deleted.

Itemset	Count
{I1, I2, I3}	3

Thus {I1, I2, I3} is frequent.

$$\begin{array}{l}
 I_3 \leftarrow \{I_1, I_2\} \\
 I_2 \leftarrow \{I_1, I_3\} \\
 I_1 \leftarrow \{I_2, I_3\}
 \end{array}$$

Step 7: Generate Association Rule :

$$\bullet \{I_1, I_2\} \rightarrow \{I_3\}$$

$$\therefore \text{Confidence} = \frac{\text{support} \{I_1, I_2, I_3\}}{\text{support} \{I_1, I_2\}} \times 100.$$

$$= \frac{3}{4} \times 100$$

$$= 75\%$$

$$\bullet \{I_1, I_3\} \Rightarrow \{I_2\}$$

$$\therefore \text{Confidence} = \frac{\text{support} \{I_1, I_2, I_3\}}{\text{support} \{I_1, I_3\}} \times 100$$

$$= \frac{3}{3} \times 100$$

$$= 100\%$$

$$\bullet \{I_2, I_3\} = \frac{3}{4} \times 100 = 75\%$$

$$\downarrow$$

$$\{I_1\}$$

$$\bullet \{I_1\} \rightarrow \{I_2, I_3\}$$

$$\therefore \text{Confidence} = \frac{\text{support} \{I_1, I_2, I_3\}}{\text{support} \{I_1\}} \times 100$$

$$= \frac{3}{4} \times 100 = 75\%$$

$$\bullet \{I_2\} \Rightarrow \{I_1, I_3\}$$

$$= \frac{3}{5} \times 100 = 60\%$$

$$\bullet \{I_3\} = \frac{3}{4} \times 100 = 75\%$$

This shows that all the above association rules are strong if min. confidence threshold is 60%.

Q] Explain FP-Growth Algorithm with its advantage and disadvantage

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- FP Growth Algorithm stands for Frequent pattern growth algorithm.
 - This algorithm is an improvement to the Apriori method.
 - A frequent pattern is generated with the need for candidate generation.
 - FP growth algorithm represents the database in the form of a tree called a frequent pattern tree or FP tree.
 - This tree structure will maintain the association betⁿ the itemsets.
 - How FP Growth works:
 1. Build FP tree: This algo. scans the dB and builds FP tree, which is a compressed representⁿ of the database.
 2. mine FP-tree: Algo. recursively mines the FP tree to generate frequent itemsets.
 - Advantages:-
 - i) Faster: Pairing not done of items here, so this makes this algo. faster.
 - ii) It is efficient & Scalable for mining both long & short frequent patterns.
 - Disadvantages:-
 - i) It is expensive
 - ii) FP Growth more complex to implement than Apriori.
 - iii) Not suitable for incremental mining, where dB is updated frequently.

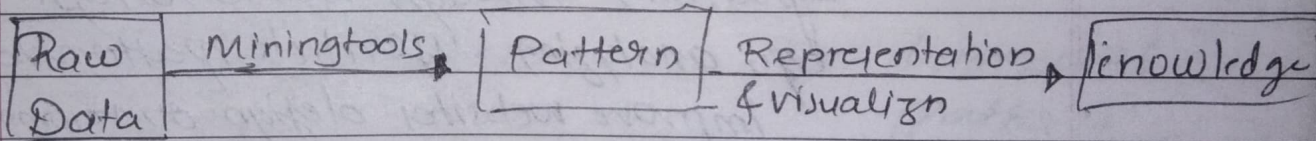
Compare and contrast data mining & Web Mining

Parameters	Data Mining	Web mining
Definition	Data Mining is the process that attempts to discover pattern & hidden knowledge in large datasets in any system.	Web Mining is the process of data mining technique to automatically discover & extract info. from web documents.
Concept	Pattern identification from data available in any system.	Pattern identification from web data.
Categories	Clustering, Classification, regression, prediction, optimization and control.	Web content Mining, Web Structure Mining, Web usage Mining.
Access	Data Mining is access data privately.	Web Mining is access data publicly.
Dynamic	Data Mining is less dynamic.	Web mining is more dynamic.
Tools	Machine learning algo.	Scrapy, PageRank, Apache logs.
Who does?	Data Scientist & Engg.	Data Scientist / Analyst & Engineers

Q. What is Web Mining? Explain the techniques of web Mining?

• Web Mining:

- Web Mining is an application of data mining techniques to find information patterns from the web data.
- Web Mining is the process of data mining techniques to automatically discover & extract information from web documents & services.
- The goal of Web mining is to look for patterns in Web data by collecting and analyzing information in order to gain insight into trends, the industry, and users to general.
- Process of Web Mining:



• Web Mining Techniques:

- Web Mining can be broadly divided into 3 diff. types

① Web content Mining:

- used for mining of useful data, info. & knowledge from web page content.
- Patterns through web crawling, real data from the web.
- eg's extracting project details from e-commerce websites.

② Web Structure Mining:

- helps to find ~~another~~ use ful knowledge or information pattern from the structure of hyperlinks.

- Patterns through linked webpages through hyperlink.

- eg's Google's Page Rank algorithm uses structure mining to rank web pages.

③ Web usage Mining:

- is used for mining the weblog records (access into. of web pages) & helps to discover the user access patterns of web pages.

- patterns through web server logs.

- eg's analyzing user clicks patterns to improve websites design or recommended products.

Q.

Differentiate between

Classification

Clustering

i) Classification is supervised learning

Clustering is unsupervised learning

ii) In Classification algo. like decision tree, Bayesian classifiers are used.

In clustering algo. like, k-mean, k-medoid expected: maximization of intra-cluster

iii) Classification has prior knowledge of classes.

Cluster doesn't have any prior knowledge of classes.

iv) eg: Classification between gender.

eg: discovery of patterns.

v) Classification requires training data to build the models.

Classification does not require training data.

vi) Classification is used in spam filtering, medical diagnosis.

Clustering is used in Customer segmentation, market analysis, etc

1) Define cluster Analysis. Give the application of cluster analysis.

Ans:- Cluster Analysis is a method used in statistics and data analysis to group objects, people or data items in such a way that the objects in the same group (called a cluster) are more similar to each other than to those in other groups. It is an unsupervised learning technique used to find natural patterns or structures in data.

Application of Cluster Analysis:-

- 1) Customer Segmentation:- Grouping customers based on purchasing behavior for targeted marketing.
- 2) Image Processing:- Grouping similar pixels for image segmentation.
- 3) Document Classification:- Grouping documents with similar topics or content.
- 4) Medical Diagnosis:- Grouping patients with similar symptoms or disease patterns.
- 5) Social Media:- It can group people with similar interest or behaviors - for example, users who like the same kind of posts.
- 6) Education:- Schools can group students based on their learning styles or performance to give better support.

4.1.2 Cluster Analysis

- Cluster Analysis in data mining means that to find out the group of objects which are similar to each other in the group but are different from the object in other groups.
- A good clustering algorithm aims to obtain clusters whose:
 - (a) The intra-cluster similarities are high. It implies that the data present inside the cluster is similar to one another.
 - (b) The inter-cluster similarity is low. It means each cluster holds data that is not similar to other data.

4.1.3 Applications of Cluster Analysis

- In many applications, clustering analysis is widely used, such as data analysis, market research, pattern recognition, and image processing.
- It assists marketers to find different groups in their client base and based on the purchasing patterns. They can characterize their customer groups.
- It helps in allocating documents on the internet for data discovery.
- Clustering is also used in tracking applications such as detection of credit card fraud.
- As a data mining function, cluster analysis serves as a tool to gain insight into the distribution of data to analyze the characteristics of each cluster.
- In terms of biology, it can be used to determine plant and animal taxonomies, categorization of genes with the same functionalities and gain insight into structure inherent to populations.
- It helps in the identification of areas of similar land that are used in an earth observation database and the identification of house groups in a city according to house type, value, and geographical location.

4.1.4 Requirements of Clustering in Data Mining

light on why clustering is